Hideaki Shimazaki

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Education

***Doctor of Philosophy*, Kyoto University** — **2004 Apr - 2007 Mar**

Department of Physics, Graduate School of Science, Kyoto University, Kyoto, Japan.
Supervisor: Prof. Shigeru Shinomoto.

***Master of Arts (Neuroscience)*, Johns Hopkins University** — **2000 Aug - 2003 Nov**

Department of Neuroscience, School of Medicine, Baltimore, MD, USA.
Supervisor: Prof. Ernst Niebur.

***Bachelor of Engineering*, Keio University** — **1996 Apr - 2000 Mar**

Department of Applied Physics and Physico-Informatics, Yokohama, Japan.

Research Experience

RIKEN Brain Science Institute, Research Scientist — **2011 Apr-Present**

Postdoctral researcher at Taro Toyoizumi lab

Massachusetts Institute of Technology, Visiting Researcher — **2009 Dec-2011 Mar**

Postdoctral researcher at Emery N. Brown lab

RIKEN Brain Science Institute, Visiting Researcher — **2007 Aug-2011 Mar**

Postdoctral researcher at Sonja Gruen lab

JSPS* Research Fellow at RIKEN Brain Science Institute — **2008 Apr-2011 Mar**

JSPS* Research Fellow at Kyoto University — **2006 Apr-2008 Mar**

* Japan Society for the Promotion of Science

Teaching Experience

Mr. Christian Donner (Bernstein Center, Berlin) — **2014 Jun - Aug**

Supervision of a summer-intern project at RIKEN Brain Science Institute. Title: A Method for Large Scale Analysis of Time-Varying Neuronal Interactions.

Ms. Safura Rashid Shomali (IPM, Tehran) — **2013 Dec - Present**

Co-supervision of Ms. Safura Rashid Shomali at the School of Cognitive Sciences (SCS) in the Institute for Research in Fundamental Sciences (IPM) for her Ph.D. thesis with Prof. Majid Nili-Ahmadabadi.

Graduate School of Science and Technology at Meiji University, Tokyo, Japan — **2013 May**

Introductory courses of Neuroscience in Japanese. Course titles: “Introduction to sensory systems, perception and neural coding studies” and “Introduction to neural decoding”.

The Institute for Research in Fundamental Sciences (IPM), Tehran, Iran — **2013 Mar**

Invited two-week lecture course titled “**Introduction to statistical models of neural spike train data**” at School of Cognitive Sciences, the Institute for Research in Fundamental Sciences (IPM), Tehran, Iran.

Awards

Japanese Neural Network Society Research Award 2009	**2009 Sep 26**
Japanese Neural Network Society Young Researcher Award 2007	**2007 Nov 15**

Research Grants and Fellowships

Excellent Young Researchers Overseas Visit Program by JSPS	**2009 Dec-2011 Mar**
Research Fellowship for Young Scientists by JSPS PD	**2008 Apr-2011 Mar**
Research Fellowship for Young Scientists by JSPS DC	**2006 Apr-2008 Mar**
Japan Student Services Organization (Fellowship)	**2004-2006**
Solo Snyder Foundation, Johns Hopkins University (Fellowship)	**2002-2004**
Murata Overseas Scholarship (Fellowship and Research Grant)	**2000-2002**

Publications

Journal publications

Shimazaki H., Amari S., Brown E. N., and Gruen S., State-space analysis of time-varying higher-order spike correlation for multiple neural spike train data. PLoS Computational Biology, 2012. 8(3): e1002385, 2012.

Shimazaki H. Analysis of multiple neural spike train data using the log-linear model. (In Japanese) The brain and neural networks, 1(4), 194-203, 2011

Shimazaki H. and Shinomoto S. Kernel bandwidth optimization in spike rate estimation. Journal of Computational Neuroscience Vol. 29 (1-2) 171-182, 2010.

Shimazaki H. and Shinomoto S. A method for selecting the bin size of a time histogram. Neural Computation, Vol. 19(6), 1503-1527, 2007.

Shimazaki H. and Niebur E. Phase transitions in multiplicative competitive processes. Physical Review E, 72(1), 011912, 2005.

Tsukada M., Aihara T., Kobayashi Y., and Shimazaki H. Spatial analysis of spike-timing-dependent LTP and LTD in the CA1 area of hippocampal slices using optical imaging, Hippocampus, 15(1), 104-109, 2005.

Kobayashi Y, Shimazaki H., Aihara T., Minoru Tsukada. Spatial distributions of hippocampal LTP/LTD induced by electrically from schaffer collaterals and stratum oriens with relative timing. (In Japanese) The brain and neural networks, 8 (2), 57-64, 2001.

Refereed conference proceedings

Shimazaki H. Single-trial estimation of stimulus and spike-history effects on time-varying ensemble spiking activity of multiple neurons: a simulation study. Journal of Physics: Conference Series (2013) 473 012009

Shimazaki H., Amari S., Brown E. N., and Gruen S. State-space Analysis on Time-varying Correlations in Parallel Spike Sequences. Proc. IEEE ICASSP, 3501-3504. 2009.

Shimazaki H. and Shinomoto S. A recipe for optimizing a time-histogram. Advances in Neural Information Processing Systems, Vol. 19, 1289-1296, 2007.

Shimazaki H. and Niebur E. Correlated multiplicative modulation in coupled oscillator systems: a model of selective attention. Progress of Theoretical Physics Supplement, No.161 336-339, 2006.

Book Chapter

Japanese translation of **Principles of Neural Science 5th ed.** Appendix F: Theoretical Approaches to Neuroscience: Examples from Single Neurons to Networks. p1567-1583, Medical Science International Inc., 2014. Original: p1601-1618, McGraw-Hill, 2012.

Selected invited talks

ELC International Meeting on "Inference, Computation, and Spin Glasses" (ICSG2013). Sapporo, Japan. Jul 2013

The 3rd Mathematical Neuroscience Workshop in School of Mathematics, Institute for Research in Fundamental Sciences (IPM). Tehran, Iran. Mar 2013

Workshop on Neural Information Flow, Kyoto University, Kyoto, Japan. Jun 2012

Neurostatistics Working Group Seminar, Dept. of Biostatistics, Harvard University, Dec 2010

IEEE ICASSP2009, Special Session on `Signal Processing for Neural Spike Trains', Taipei, Taiwan. Apr 2009

NIPS 2008, Workshop on `Statistical Analysis and Modeling of Response Dependencies in Neural Populations', Whistler, Canada. Dec 2008

Riken Brain Science Institute Forums, Wako, Japan. Host: Dr. Sonja Gruen. Apr 2007

The Boadian Seminar at Mind/Brain Institute, Johns Hopkins University. Baltimore, USA. Host: Dr. Ernst Niebur. Mar 2007

Academic service

Journals reviewed for: PLOS Computational Biology (3 times), Neural Networks (3), Springer Encyclopedia (3), Computational Intelligence and Neuroscience (3), Journal of Computational Neuroscience, Neural Computation, Journal of Neuroscience Methods, Physical Review E, PLOS ONE, Cognitive Neurodynamics, Applied Mathematical Modeling.

Proceedings reviewed for: IEEE ICASSP (4 times), ICONIP, Journal of Physics: Conference Series, Progress of Theoretical Physics.

References

Prof Shigeru Shinomoto PhD supervisor

Department of Physics, Graduate School of Science, Kyoto University
Email: shinomoto@scphys.kyoto-u.ac.jp Tel: 075-753-3778

Prof Ernst Niebur MA supervisor

Department of Neuroscience, Johns Hopkins University
Email: niebur@jhu.edu Tel: 075-753-3778

Prof Sonja Grün Postdoc supervisor

Research Center Juelich, Juelich, Germany / RTWH Aachen University, Aachen, Germany
Email: s.gruen@fz-juelich.de Tel: +49-(0)2461-61-4748

Prof Emery N. Brown Postdoc supervisor

Massachusetts Institute of Technology / Warren M. Zapol Professor of Anesthesia, Harvard Medical School
Email:enb@neurostat.mit.edu Tel: +1-617-324-1879

Prof Taro Toyoizumi Postdoc supervisor

RIKEN Brain Science Institute, Saitama, Japan
Email: taro.toyoizumi@brain.riken.jp Tel: +81-48-467-9644

List of all presentations

■ 2014

Sharp T., Shimazaki H., Isomura Y., Fukai T. Behaviour- and layer-dependent synchrony in motor cortex during volitional arm movement, Neuro2014, Yokohama, Japan. Sep 11 **POSTER**

Shomali S.R., Ahmadabadi M.N., Shimazaki H., Rasuli N. S. Theoretical study on spike-timing probability in a pair of pre-post synaptic neurons. Neuro2014, Yokohama, Japan. Sep 11 **POSTER**

MaBouDi H., Shimazaki H., Abouzari M., Soltanian-Zadeh H., Amari S. Bimodal Distributions of Local Phase Variables in Natural Images. The 2014 Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, Florida. May 18 **POSTER**

MaBouDi H., Shimazaki H., Abouzari M., Amari S, Soltanian-Zadeh H. Statistical inference for directed phase coupling in neural oscillators. Computational and Systems Neuroscience (Cosyne) 2014. Salt Lake City, USA. Feb 27 **POSTER**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyoizumi T. Structured higher-order interactions explain the simultaneous silence of neural populations. Mechanism of Brain and Mind, Rusutsu, Hokkaido, Japan Jan 8-10 **POSTER**

■ 2013

Shimazaki H., Sadeghi K., Ikegaya Y., Toyoizumi T., The simultaneous silence of neurons explains structured higher-order interactions in ensemble spiking activity, SfN2013. San Diego, USA. Nov 9-13 **POSTER**

Shimazaki H., State-space Analysis of Time-varying Higher-order Interactions: its Applications to Neuroscience, ELC International Meeting on "Inference, Computation, and Spin Glasses" (ICSG2013). Sapporo, Japan. Jul 28 **INVITED TALK**

Shimazaki H., Higher-order interactions in population activity of hippocampal CA3 neurons, Workshop on statistical analysis of neurophysiological and clinical data, Kyoto, Japan. Jul 8-9 **TALK**

Shimazaki H., Estimating Time-varying Higher-order Neuronal Interactions in Awake Behaving Animals, Modeling Neural Activity: Statistics, Dynamical Systems, and Networks, Lihue, Hawaii, USA. 2013 June 26-28 **TALK**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyoizumi T., The simultaneous silence of neurons explains higher-order interactions in ensemble spiking activity, Neuro2013, Kyoto, Japan. Jun. 20 **POSTER**

Shimazaki H. Estimating dynamic neural interactions in awake behaving animals. The 3rd Mathematical Neuroscience Workshop in School of Mathematics, Institute for Research in Fundamental Sciences (IPM). Tehran, Iran. Mar. 13 **INVITED TALK**

Shimazaki H. The Simultaneous Silence of Neurons Explains Higher-Order Interactions in Ensemble Spiking Activity. The 3rd Mathematical Neuroscience Workshop in School of Mathematics, Institute for Research in Fundamental Sciences (IPM). Tehran, Iran. Mar. 13 **INVITED TALK**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyoizumi T. The simultaneous silence of neurons explains structured higher-order interactions in spontaneous spiking activity. Dialog between Neuroscience and Statistics, Institute of Mathematics and Statistics, Tokyo, Japan Feb 18-19 **TALK**

■ 2012

Shimazaki H., Sadeghi K., Ikegaya Y., Toyoizumi T. Joint inactivation statistics of population spiking activities. RIKEN BSI Retreat 2012, Nov 12-13, Karuizawa, Japan **POSTER**

Shimazaki H. The simultaneous silence of neurons explains structured higher-order interactions in ensemble spiking activity. RIKEN BSI Lunch Seminar 2012 Nov. 8 **TALK**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyoizumi T. Joint inactivation statistics of population spiking activities. Workshop on statistical aspects of neural coding. Nov 1-2. Kyoto University & Ritsumeikan University. **TALK**

Shimazaki H. Tracking Dynamic Neural Interactions in Awake Behaving Animals. Workshop on neural information flow, Kyoto University, Kyoto Jun 20 2012 **INVITED TALK**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyoizumi T. The simultaneous silence of neurons explains higher-order interactions in ensemble spiking activity. Computational and Systems Neuroscience (Cosyne) 2012. Salt Lake City, USA. 2012 Feb 23-26 **POSTER (Reviewed)**

■ 2011

Shimazaki H., Amari S., Brown E. N., and Gruen S. Dynamics of Higher-order Spike Correlation in an Awake Behaving Monkey: Analysis by a State-space Model. RIKEN BSI Retreat, Karuizawa, Japan. Oct 31 **POSTER**

Shimazaki H., Ikegaya Y., and Toyoizumi T. A New Sparse Log-linear Model for Simultaneously Active and Inactive Neurons. RIKEN BSI Retreat, Karuizawa, Japan. Oct 31 **POSTER**

Shimazaki H. and Brown E.N. Copula-based Mixture Time-series Model of Continuous and Point Processes for Synthetic Analysis of Neural Signals. RIKEN BSI Retreat, Karuizawa, Japan. Oct 31 **POSTER**

Shimazaki H. and Brown E. N. Constructing a joint time-series model of continuous and Bernoulli/Poisson processes using a copula Computational and Systems Neuroscience 2011 Salt Lake City, USA. Feb 24-27 **POSTER (Reviewed)**

■ 2010

Shimazaki H. Analysis of Dynamic Neural Spike Data: From Firing Rates to Spike Correlations. Neurostatistics Working Group Seminar, Dept. of Biostatistics, Harvard University, Dec 1 **INVITED TALK**

Shimazaki H. Detection of dynamic cell assemblies by the Bayes Factor. Workshop on spatio-temporal neuronal computation, Kyoto University, Japan Sep 6-7 **INVITED TALK**

Shinomoto S*, Shimazaki H, and Shimokawa T. Characterizing neuronal firing with the rate and the irregularity. Neuro2010, Kobe, Japan Sep. 2 S3-10-1-3 **TALK**

Shimazaki H., Gruen S., and Amari S. Analysis of subsets of higher-order correlated neurons based on marginal correlation coordinates. Cosyne 2010, Salt Lake City, USA. 1-63 Feb 25-28 **POSTER (Reviewed)**

■ 2009

Shimazaki H., Amari S., Brown E. N., and Gruen S. State-space Model of Dynamic Spike Correlation. Japanese Neural Network Society 2009. Sendai, Japan. Sept. 24-26. TALK+POSTER **Japanese Neural Network Society 2009 Distinguished Research Award**

Shimazaki H. Analysis of Dynamic Spike Data: From Spike Rate to Multiple Neuron Spike Correlation. The 6th Brain Lunch Seminar. RIKEN Brain Science Institute, Saitama, Japan. Sep 8. **TALK**

Shimazaki H., Amari S., Brown E. N., and Gruen S. Bayes Factor Analysis for Detection of Time-dependent Higher-order Spike Correlations. CNS2009. Berlin, Germany. Jul 18-23 P99 **POSTER**

Shimazaki H. and Shinomoto S. Histogram binwidth and kernel bandwidth selection for the Spike-rate estimation. CNS2009. Berlin, Germany. Jul 18-23 P116 **POSTER**

Shimazaki H., Amari S., Brown E. N., and Gruen S. Estimating time-varying spike correlations from parallel spike sequences, German-Japanese Workshop "Computational and Systems Neuroscience", Berlin, May 25-28, 2009 **POSTER**

Shimazaki H., Amari S., Brown E. N., and Gruen S. State-space Analysis on Time-varying Correlations in Parallel Spike Sequences. IEEE ICASSP2009 Special Session on `Signal Processing for Neural Spike Trains', Taipei, Taiwan. Apr 24 SS-L10.4 **INVITED LECTURE**

Shimazaki H., Amari S., Brown E. N., and Gruen S. Detection of non-stationary higher-order spike correlation. Cosyne 2009, Salt Lake City, USA. Feb 26 -Mar 3 II-62 **POSTER (Reviewed)**

■ 2008

Shimazaki H., Amari S., Brown E. N., and Gruen S. State-space Analysis on Time-varying Higher-order Spike Correlations. NIPS2008 Workshop on `Statistical Analysis and Modeling of Response Dependencies in Neural Populations', Whistler, Canada Dec 13, 17:00-17:30 **INVITED TALK**

Shimazaki H. and Gruen S. Selecting a state-space model of higher-order correlations in parallel spike trains from competing hierarchical log-linear models. RIKEN BSI Retreat 2008, Karuizawa, Japan. Nov 4-5 **POSTER**

Shimazaki H. and Shinomoto S. Spike-rate Estimation with Locally Adaptive Kernel Method, Sep 24-26 Tsukuba (In Japanese) PS3-4 **SPOTLIGHT POSTER**

Shimazaki H. and Gruen S. Estimating time-dependent higher-order interactions in parallel spike trains. Neuro2008, Tokyo, Japan. Jul 9-11 **POSTER**

Shimazaki H., Brown E. N. and Gruen S. State-space Analysis on Time-dependent Correlation in Parallel Spike Trains. Statistical Analysis of Neuronal Data (SAND4), Pittsburgh, PA, USA. May 29-31 **POSTER**

■ 2007

Shimazaki H. and Gruen S. Estimation of Time-dependent Higher-Order Interactions in Parallel Spike Trains. Riken BSI Retreat, Karuizawa, Japan. Nov 26-28 **POSTER**

Shimazaki H. and Shinomoto S. Kernel Width Optimization in the Spike-rate Estimation. Neural Coding 2007, Montevideo, Uruguay. Nov 7 POSTER **BEST POSTER AWARD**

Shimazaki H. and Shinomoto S. Optimization of a Histogram of Spike Data. Neuro2007, Yokohama, Japan. Sep 11 P2-k22 **POSTER**

Shimazaki H. A Recipe for Optimizing a Time Histogram of Spike Data. Riken BSI Forums at RIKEN Brain Science Institute, Wako, Japan. Host: Sonja Gruen Apr 17 **INVITED TALK**

Shimazaki H. A recipe for constructing a Peri-stimulus Time Histogram. The Boadian Seminar at Mind/Brain Institute, Johns Hopkins University. Host: Ernst Niebur. Mar 1 **INVITED TALK**

Shimazaki H. and Shinomoto S. A recipe for optimizing a time-histogram with variable bin sizes. Computational and Systems Neuroscience 2007, Salt Lake City, Feb 22-27 **POSTER (Reviewed)**

■ 2006

Shimazaki H. and Shinomoto, S. A recipe for optimizing a time-histogram. Neural Information Processing Systems Dec 4-9, Whistler, B. C. **POSTER (Reviewed) selected as a SPOTLIGHT POSTER**

Shimazaki H. A method to optimize a time histogram by correcting onset latencies of spike sequences Japanese Neural Network Society 2006 Sep 1-21 Nagoya, Japan (In Japanese) O4-1 TALK＋POSTER **Japanese Neural Network Society 2007 Young Researcher Award**

Shimazaki H. Self-organized criticality by natural selection. Frontiers in Dynamics: Physical and Biological Systems Tokyo, Japan, May 22-24, 2006, **POSTER**

Shimazaki H. and Shinomoto S. Recipes for constructing an optimal time histogram. Statistical Analysis of Neuronal Data (SAND3) Pittsburgh, PA, May 12-13, 2006, **POSTER**

■ 2005

Shimazaki H. and Shinomoto S. A Recipe for Making a Time Histogram with an Optimal Bin Width from Spike Sequences. Japanese Neural Network Society 2005, Kagoshima (In Japanese) **SPOTLIGHT POSTER**